

Deploying a Node.js Application on Kubernetes using Minikube (Step-by-Step Guide)

This guide explains how to deploy a Node.js application into a local Kubernetes cluster using Minikube. It includes building a Docker image, loading it into Minikube, creating Kubernetes deployments and services, and accessing the application from the browser.

1. Prerequisites

Make sure the following tools are installed:

- 1 Docker
- 2 kubectl (Kubernetes CLI)
- 3 Minikube

Verify Docker:

```
docker --version
docker ps
```

Verify kubectl:

```
kubectl version --client
```

2. Install Minikube

```
wget https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
sudo install minikube-linux-amd64 /usr/local/bin/minikube
minikube version
```

3. Start Kubernetes Cluster

```
minikube start --driver=docker
kubectl get nodes
```

4. Build Node.js Docker Image

```
npm install
docker build -t my-k8s-app:v1 .
```

Verify image:

```
docker images
```

5. Tag Docker Image

```
docker tag my-k8s-app:v1 dolendra/my-k8s-app:v1
```

6. Load Image into Minikube

```
minikube image load dolendra/my-k8s-app:v1
```

7. Create Kubernetes Deployment

deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-k8s-app-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: my-k8s-app
  template:
    metadata:
      labels:
        app: my-k8s-app
    spec:
      containers:
        - name: my-k8s-app
          image: dolendra/my-k8s-app:v1
          ports:
            - containerPort: 8080
kubectl apply -f k8s/deployment.yaml
```

8. Create Kubernetes Service

service.yaml:

```
apiVersion: v1
kind: Service
metadata:
  name: my-k8s-app-service
spec:
  type: NodePort
  selector:
    app: my-k8s-app
  ports:
    - port: 8080
      targetPort: 8080
      nodePort: 32000
kubectl apply -f k8s/service.yaml
```

9. Verify Deployment

```
kubectl get pods
kubectl get deployments
kubectl get services
kubectl get all
```

10. Access Application

```
minikube service my-k8s-app-service
```

11. Debugging Commands

```
kubectl logs <pod-name>
kubectl describe pod <pod-name>
```

12. Scale Application

```
kubectl scale deployment my-k8s-app-deployment --replicas=4
kubectl get pods
```

13. Stop or Delete Resources

```
kubectl delete service my-k8s-app-service
```

```
kubectl delete deployment my-k8s-app-deployment
```

```
minikube stop
```

```
minikube start --driver=docker
```

```
minikube delete
```